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A Additional Related Work

The problem of linear concept erasure is an instance of the general problem of information removal. Information removal methods generally divide into adversarial methods, which are applied during training, and the post-hoc linear methods considered in this paper. Adversarial methods [8, 42, 4, 9, 44] use a gradient-reversal layer during training to induce embeddings that do not encode the protected attribute. However, Elazar and Goldberg [9] have shown that these methods fail in exhaustively removing all the information associated with the protected attribute: it is often possible to train new adversaries that successfully recover the removed information. Linear methods have been proposed as a tractable alternative, where one identifies a linear subspace that captures the concept of interest, and neutralizes it using algebraic techniques. Different methods have been proposed for the identification of the subspace, e.g. PCA and variants thereof [3, 20], orthogonal-rotation [7], classification-based [29], spectral [36, 37] and adversarial approaches [31].

Few works theoretically characterize the condition of linear guardedness. Haghighatkhah et al. [16] extensively analyzed the problem of preventing linear classification, with the focus on decreasing accuracy. They provide a constructive proof of an optimal intervention for an SVM classifier. Ravfogel et al. [33] have proposed a formal definition of linear guardedness based on $\mathcal{V}$ information, and characterized the fairness implications of guardedness; we show the relations with our definition above. Ravfogel et al. [31] provide an adversarial formulation of the problem, derive a closed-form solution to certain cases, and propose an SGD-based optimization for others. While they seek an *orthogonal* projection, we empirically showed that their solution is very close to ours. Sadeghi et al. [35] and Sadeghi and Boddeti [34] both study an adversarial formulation of concept erasure for linear regression, and they trade-off with main-task performance. In contrast to Ravfogel et al. [31], they consider a general linear adversary, i.e. not necessarily a projection matrix. Closest to our work are Kleindessner et al. [20], Haghighatkhah et al. [17], Shao et al. [36]. As we showed above (§ 4), those methods do achieve the goal of linear guardedness though they are unable to prove this fact. At the same time, they are not optimal in terms of damage to the original embedding space.

B Equivalence of Guardedness with the Optimality of Constant Predictors

The following two theorems establish the equivalence of conditions 1 and 2 (indeed, they do so in the general setting, with no assumption of convex loss or linear predictors).

Theorem B.1. *Suppose X ($\mathcal{V}, \mathcal{Z}$)-guards Z . Then for every loss $\mathcal{L} \in \mathcal{L}$, the corresponding trivially attainable loss $L_\tau^{(\mathcal{Z}, \mathcal{L})}$ cannot be improved upon by any predictor $\eta(\cdot; \theta) \in \mathcal{V}$, i.e. $L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$.*

Proof. Consider the null random vector $X'(\omega) = \mathbf{0}$. Since all predictors are constant on X' , and the trivially attainable loss gives the *best* available expected loss among constant predictors, we must have:

$$L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X'; \theta), Z)] \quad (4)$$

The right side of equation (4) is the best possible loss achievable by a function $\eta(\cdot; \theta)$ on the joint distribution of (X', Z) , which by the definition of guardedness is upper bounded by the best possible loss achievable on the joint distribution of (X, Z) :

$$\inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X'; \theta), Z)] \leq \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)] \quad (5)$$

Combining equations (4) and (5), and the fact that all constant functions exist in our function class $\mathcal{V} = \{\eta(\cdot; \theta)\}$, we arrive at our desired result:

$$L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$$

□

Theorem B.2. *Suppose that for every loss $\mathcal{L} \in \mathcal{L}$, the corresponding trivially attainable loss $L_\tau^{(\mathcal{Z}, \mathcal{L})}$ cannot be improved upon by any predictor $\eta(\cdot; \theta) \in \mathcal{V}$, i.e. $L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$. Then X ($\mathcal{V}, \mathcal{Z}$)-guards Z .*

Proof. Let $X' : \Omega \rightarrow \mathbb{R}^d$ be any other random data vector with finite first moment.

Since all constant predictors exist in our predictor class $\mathcal{V} = \{\eta(\cdot; \theta)\}$, the best loss achievable on (X', Z) by functions in $\mathcal{V}$ must be at least as good as the trivially attainable loss (the best loss available by such constant predictors):

$$\inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X'; \theta), Z)] \leq L_{\tau}$$

By assumption, the trivially attainable loss cannot be improved upon over (X, Z) by predictors in $\mathcal{V}$:

$$L_{\tau} = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$$

Since our choice of X' was arbitrary, this shows that X maximizes the minimal achievable loss, so X $(\mathcal{V}, \mathcal{L})$ -guards Z . $\square$

C Linear Guardedness is Equivalent to Linear Statistical Parity

To measure the effect of linear guardedness on main-task classifiers, we use the following minimal definition of “fairness” with respect to an attribute, adapted from Edwards and Storkey [8].

Definition C.1 (Statistical Parity). *Let X and Z be defined as above, and let f be a function with domain $\mathbb{R}^d$. Then f exhibits **statistical parity** with respect to Z when evaluated on X if*

$$\forall z \in \mathcal{Z} : \mathbb{E}[f(X)|Z = z] = \mathbb{E}[f(X)].$$

We now prove the equivalence of conditions 3 and 5.

Theorem C.2. *Let X and Z be defined as above. Then every linear predictor $f(\mathbf{x}) = \mathbf{b} + \mathbf{W}\mathbf{x}$ exhibits statistical parity w.r.t. Z when evaluated on X if and only if each class-conditional mean $\mathbb{E}[X|Z = z]$ is equal to $\mathbb{E}[X]$.*

Proof. Suppose each class-conditional mean $\mathbb{E}[X|Z = z]$ is equal to $\mathbb{E}[X]$. Then by the linearity of expectation, we have for all $z \in \mathcal{Z}$:

$$\mathbb{E}[f(X)|Z = z] = \mathbb{E}[\mathbf{W}X + \mathbf{b}|Z = z] = \mathbf{W}\mathbb{E}[X|Z = z] + \mathbf{b} = \mathbf{W}\mathbb{E}[X] + \mathbf{b} = \mathbb{E}[f(X)].$$

This matches the definition of statistical parity provided in Definition C.1.

Conversely, suppose every linear predictor $f(\mathbf{x}) = \mathbf{b} + \mathbf{W}\mathbf{x}$ exhibits statistical parity w.r.t. Z when evaluated on X . Then this holds for the identity function $\text{id}(\mathbf{x}) = \mathbf{x}$, and thus for all $z \in \mathcal{Z}$:

$$\mathbb{E}[X|Z = z] = \mathbb{E}[\text{id}(X)|Z = z] = \mathbb{E}[\text{id}(X)] = \mathbb{E}[X].$$

$\square$

D Implications for Prior Work

In this section we discuss the implications of Theorem 4.1, which characterizes the necessary and sufficient conditions for an affine erasure function to yield a perfectly linearly guarded dataset, for methods proposed in prior work.

Spectral Attribute Removal (SAL) [36] uses the top n left singular vectors of Σ_{XZ} to construct an orthogonal projection matrix $\mathbf{Q}_{\text{SAL}} = \mathbf{I} - \mathbf{U}_{:n} \mathbf{U}_{:n}^T$ which is then applied to X . Notably, while n is presented as a free parameter in their method, all of their experiments involve binary classification problems where Z is a one-hot vector, and n is set to a value no greater than 2. We’ll call the version of SAL where $n = \text{rank}(\Sigma_{XZ})$, “full-rank SAL.” Since these left singular vectors are an orthonormal basis for Σ_{XZ} ’s column space, Theorem 4.1 implies that full-rank SAL guarantees linear guardedness.

Mean Projection (MP) [17] orthogonally projects X onto the orthogonal complement of the span of the difference in class centroids $\mathbb{E}[X|Z = 1] - \mathbb{E}[X|Z = 0]$, where Z is assumed to be binary. Since the centroids are equal after the projection, this method guarantees linear guardedness by Theorem 3.1. In fact, by Theorem 3.4, MP is mathematically equivalent to SAL when Z is a one-dimensional random vector taking one of two possible values.